

**DESIGN AND IMPLEMENTATION OF A REAL-TIME LAN-BASED QUEUING
MANAGEMENT AND STATISTICAL REPORTING SYSTEM FOR THE
BUSINESS PERMITS, LICENSING - LOCAL INVESTMENT AND
ENTERPRISE OFFICE (BPL-LIEO) OF LGU ROSARIO, LA UNION**

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APPROVAL SHEET

This thesis proposal, **Design and Implementation of a Real-Time LAN-Based Queuing Management and Statistical Reporting System for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) of LGU Rosario, La Union**, prepared and submitted by **Rojean F. Tandas, Manrijan M. Barbosa, Carren Nicole A. Estrelles, John David R. Garcia, Jade Allysa V. Rulloda**, in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science was examined and passed on _____ by the Thesis Committee composed of:

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RFT
MMB
CNAE
JDRG
JAVR

DEDICATION

This study is dedicated to our families for their endless love, encouragement, and sacrifices that have given us the strength to complete this journey.

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TABLE OF CONTENTS

	Page
TITLE PAGE	i
APPROVAL SHEET	ii
ACKNOWLEDGMENTS	iii
DEDICATION	v
LIST OF TABLES	viii
LIST OF FIGURES	ix
LIST OF PLATES	ix
ABSTRACT	xi

CHAPTER

1 INTRODUCTION

Situation Analysis	1
Statement of Objectives	11
Time and Place of Study	12
Definition of Terms	13

2 METHODOLOGY

Research Design	16
Materials and Procedures	16
Hardware	16
Software	17
Software Development Life Cycle (SDLC)	17

System Testing Procedures	18
Data Gathered	22
Data Analysis	23

LITERATURE CITED

APPENDICES

A Letter to the Adviser	28
B Interview Guide Questions	29
C Evaluation Questionnaire.....	30
D Letter to the Panel Members.....	33

CURRICULUM VITAE

Rojean F. Tandas	35
Carren Nicole A. Estrelles	37
Jade Allysa V. Rulloda	39
John David R. Garcia.....	41
Manrijan M. Barbosa	43

LIST OF TABLES

Table No		Page
1	Table 1. Distribution of Respondents	21

LIST OF FIGURES

Figure No.		Page
1	Figure 1. Custom Agile Software Development Life Cycle Model for the BPL-LIEO Queuing System	18

LIST OF PLATES

Plates No		Page
1	Title of Plate 1	43
2	Title of Plate 2	43

ABSTRACT

ROJEAN F. TANDAS, MANRIJAN M. BARBOSA, CARREN NICOLE A. ESTRELLES, JOHN DAVID R. GARCIA, JADE ALLYSA V. RULLODA (2027). **Design and Implementation of a Real-Time LAN-Based Queuing Management and Statistical Reporting System for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) of LGU Rosario, La Union.** Bachelor of Science in Computer Science. Don Mariano Marcos Memorial State University South La Union Campus, College of Computer Science, Agoo, La Union.

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Many government offices continue to rely on manual queuing methods resulting in long waiting times, disorganized client flow, and delays in service delivery. These challenges are particularly evident in the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) where a high volume of clients is accommodated daily for transactions such as business registration and permit renewal.

This study proposes the design and implementation of a Real-Time LAN-Based Queuing Management and Statistical Reporting System for the BPL-LIEO. The proposed system aims to reduce waiting time, improve queue management, enhance operational efficiency, and provide accurate statistical reports to support better service delivery for both clients and BPL-LIEO staff. The system is designed to automate queue management by generating queue numbers, displaying the current and next queue numbers in real time across the local area network (LAN), monitoring queue statuses (Waiting, Serving, Completed, and Skipped), and enabling staff to efficiently call, skip, remove, and complete queue numbers. The system also provides a statistical reporting module that records and summarizes queue transactions, allowing BPL-LIEO personnel to monitor daily operations and generate reports for decision-making and performance evaluation.

The study will use developmental and descriptive research designs with the use of Agile methodology for system development. The developed system will be evaluated based on the ISO/IEC 25010 Software Quality Model to assess its functional suitability, usability, reliability, performance efficiency, security, maintainability, and portability.

Keywords: *Real-Time LAN-Based Queuing Management System, Statistical Reporting System, Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO), Queue Monitoring, Service Automation*

Chapter 1

INTRODUCTION

Situation Analysis

Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO)s (BPL-LIEOs) play an important role in providing government services to individuals and business owners who apply for new business permits, renew existing permits, or process other business-related transactions. The increasing number of clients during business registration and renewal periods requires an organized process for receiving applications, checking requirements, assigning clients to the appropriate service process, and monitoring the progress of transactions. When queuing and client movement are not properly coordinated, clients may experience uncertainty about their position in the queue, while personnel may experience difficulty in monitoring and managing multiple transactions at the same time. The proposed study focuses on the development of an automated queue management system featuring real-time status updates and staff-controlled queue operations for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO).

Queue management has been studied as an important component of service systems because the way clients are assigned to available service personnel can directly affect waiting time and service efficiency. Ainbinder, Temnikov, and Allalouf (2024) compared global and local queue structures in systems with workers having different processing rates. The study examined how queue architecture and routing policies affect waiting time and workload distribution. The findings emphasized that queue design and the manner in which

tasks are distributed among service personnel can influence the efficiency of a service system. This is relevant to the present study because BPL-LIEO operations may involve personnel performing different functions, making appropriate queue assignment and monitoring important. The study can be adopted conceptually by considering how queue information is organized and how clients are directed to available personnel or service windows.

Lee et al. (2022) examined a priority queueing system involving multiple types of customers and different levels of service priority. The study proposed a flexible priority scheme in which service resources could be allocated according to the current queue condition. The researchers emphasized that priority mechanisms can be used to improve resource allocation while maintaining a degree of fairness among different customer groups. This is relevant to the present study because the proposed BPL-LIEO system includes a Priority Waiting status for clients who return after being placed in a Return Tomorrow state. The concept of priority queueing can therefore help support the design of a system where authorized personnel can manage exceptional queue conditions without losing control of the overall client flow.

These international studies demonstrate that queue management is not limited to issuing numbers. Queue architecture, resource allocation, service priorities, and the distribution of clients among available personnel can affect waiting time and operational efficiency. The present study adopts these concepts at a simpler and more practical level by providing BPL-LIEO personnel with real-time queue information and staff-controlled functions for calling, completing, skipping, removing, and managing returning clients.

In Asia, queue management has also been applied through web-based and real-time technologies. Chantima and Pochan (2023), from Naresuan University in Thailand, developed a mathematical model for real-time queue management based on equality principles. The researchers compared a conventional first-come-first-served queue with an appointment-based real-time queue and used Microsoft Excel with Visual Basic for Applications (VBA) to analyze waiting times. Their study demonstrated the importance of continuously considering queue conditions when managing client waiting time. This study is useful to the present research because it supports the idea that queue information should be updated according to the current state of the service process rather than relying only on a static queue sequence.

Santoso et al. (2024), from Politeknik Caltex Riau in Indonesia, developed a queue management system for Puskesmas Umban Sari. The system was designed as an Android- and website-based online queueing system to address difficulties in managing patient queues. It included registration and data-recapitulation functions and supported both priority and linear queues. The system was developed using the Rational Unified Process and evaluated through black-box, usability, and performance testing. This study is relevant to the present research because it demonstrates how different queue types can be incorporated into a computerized system and how queue information can be supported by registration and record management. The present study can adopt the concept of priority and ordinary queues while applying them to BPL-LIEO transactions rather than healthcare services.

Fauzi and Suhaimi (2026) developed and evaluated QKlinik, a web-based queue management system for outpatient clinics in Malaysia. The system incorporated online

queue-number reservation, QR-code access, real-time waiting-status tracking, and an integrated interface for clinic-related services. The study addressed the need for greater service transparency and improved patient flow through digital queue information. The system is relevant to the present study because it demonstrates the practical utility of web-based, real-time queue monitoring and public queue status visibility. These concepts directly support the proposed BPL-LIEO system's architectural components, specifically the deployment of a dedicated ticket generation interface, a staff-controlled operation dashboard, and a centralized real-time monitor display for transparent client queue tracking.

The Asian studies reviewed indicate that real-time information, priority management, web-based access, and flexible queue structures can improve the organization of service processes. However, these studies were conducted mainly in healthcare and other service environments. The present study applies similar technological concepts to the specific workflow of a local government Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO), particularly for new business permit applications, renewals, and other supported transactions.

The Philippine government has increasingly promoted the digitalization of business permitting and other frontline government services. In 2024, the Anti-Red Tape Authority (ARTA), together with the Department of the Interior and Local Government, Department of Information and Communications Technology, and Department of Trade and Industry, intensified efforts to support local government units in establishing electronic Business One-Stop Shops (eBOSS). ARTA emphasized that eBOSS is intended to provide more efficient filing of applications and issuance of local business licenses, permits, clearances, and authorizations.

The scale of this national transition is also reflected in reports from the Philippine News Agency. In January 2024, ARTA reported that 630 out of 1,637 LGUs had either fully or partially implemented eBOSS and that assistance would be provided to LGUs to support wider implementation. By February 2024, ARTA was targeting an additional 120 LGUs for full eBOSS compliance. These developments demonstrate the continuing national movement toward digital business permitting and the need for local governments to modernize their service processes.

Dela Merced, Balmeo, and De Castro (2024) examined the client perspective of the Electronic Business One-Stop Shop (eBOSS) among small and medium enterprises in Quezon City. Their study investigated the relationship between the implementation of eBOSS and client satisfaction and identified opportunities for improving the electronic platform. This study is useful to the present research because it demonstrates that digital business permitting should be examined not only from the perspective of government personnel but also from the perspective of clients who use the service. The present study similarly considers both clients and BPL-LIEO personnel as intended users of the proposed queue management system.

Espinosa, Serrano, and Balmes (2024) developed an automated enrollment queueing system in a Philippine university to improve waiting-line management and operational efficiency. The study examined the weaknesses of the existing enrollment process and developed a computerized queueing system intended to reduce overcrowding, minimize waiting-related difficulties, reduce employee workload, and improve service delivery. Although the setting was an educational institution rather than a government office, the

study provides a relevant Philippine example of how automation can organize client flow and reduce manual queue management.

Alegado and Himo (2022) developed a Community Request Queue Management System for the Local Government Unit of Cabanatuan City. The study addressed difficulties involving manual storage of information, retrieval of records, and service arrangements within a local government setting. The system followed the stages of requirements analysis, system design, development, testing, deployment, and maintenance. This study is particularly relevant because it demonstrates the application of a computerized queue and records-management approach in a Philippine local government environment. The present study extends this concept by focusing specifically on business permit-related transactions and real-time staff-controlled queue operations.

The national literature shows that Philippine government offices are increasingly moving toward electronic processing and automated service management. However, electronic business permitting and queue management are not exactly the same function. An eBOSS primarily supports digital business permit processing, while the proposed study focuses on organizing the physical client flow and service sequence within a BPL-LIEO. Therefore, the proposed system seeks to complement digital government initiatives by providing a queue management layer that can organize clients, provide real-time queue information, and give staff direct control over queue operations.

At the regional level, a particularly relevant study was conducted by Rodriguez et al. (2025) in Alaminos City, Pangasinan which is part of Region I. The researchers examined the development and implementation of an advanced queuing system for the Local Government Unit of Alaminos City. The study aimed to streamline government service

processes, minimize waiting time, and improve service efficiency and customer experience. Using a mixed-methods approach involving waiting-time analysis, system-performance measures, customer satisfaction surveys, and staff interviews, the researchers reported improvements in waiting time, customer satisfaction, workflow efficiency, and the organization and transparency of the queueing process. The system also allowed personnel to manage transactions and prioritize urgent cases.

The Alaminos City study is particularly significant to the present research because it demonstrates that queue management is already being explored within a local government setting in Region I. Its use of staff interviews and waiting-time analysis also provides a useful approach for the present study's requirements-gathering process. The present research can adopt the concept of combining client feedback and staff requirements when designing a queue management system. However, the proposed study differs in its specific focus on the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) of Rosario, La Union and its inclusion of transaction-based tickets, real-time queue status, return-tomorrow handling, and staff-controlled queue operations.

Regional government developments show continuing digitalization of business permit services. In 2024, the Anti-Red Tape Authority recognized Bacnotan, La Union for fully complying with the eBOSS system. The municipality was reported as the second LGU in the Ilocos Region, after the City of San Fernando, La Union, to pass ARTA's assessment of fully streamlined and digitalized business permit application-to-issuance processes. The eBOSS implementation included a unified application form, electronic tax bills or orders of payment, online uploading of requirements, electronic permits, integrated online payment, and courier integration.

These developments indicate that digital business permit processing and queue-related service improvements are already emerging within Region I. Nevertheless, the existence of digital permit systems does not mean that every municipality uses the same workflow or queue management mechanism. The present study therefore focuses on the actual operational requirements of the selected BPL-LIEO in Rosario, La Union rather than assuming that a system used by another LGU can be directly applied to the selected client. Within La Union, local government units have implemented different approaches to business permit processing. In Agoo, La Union, the Municipal Government organized its 2025 Business One-Stop Shop through the Municipal Treasury Office and Business Permit and Licensing Office. The BOSS brought representatives from different LGU offices and agencies, including the Bureau of Fire Protection, Sangguniang Bayan Office, Municipal Engineering Office, and Agoo Public Safety and Order Office, into one location to facilitate the processing of business permit applications and renewals. The arrangement demonstrates the importance of coordinating several offices and service functions during business permit processing.

The Municipality of Sudipen, La Union also published its 2025 Business One-Stop Shop procedures. The municipality distinguishes requirements for new business applications from renewal transactions. For new businesses, requirements include proof of registration, documents for tax computation, and other applicable permits and clearances. Renewal transactions have different requirements, including the basis for computing business taxes and other supporting documents. The published process instructs clients to submit documents and application forms and wait for the corresponding assessment before payment. This distinction between new and renewal transactions supports the present

study's observation that transaction types should be identified at the beginning of the queue process because they may require different requirements and processing procedures.

The Provincial Government of La Union likewise publishes separate requirements and procedures for new and renewal business registration. Its service procedure identifies the submission and review of forms and documents, verification, assessment of current amounts or previous balances, and forwarding of assessment information for payment. This demonstrates that business-related government transactions involve several stages and require proper coordination between clients and personnel.

These local examples demonstrate that business permit processing in La Union involves different transaction types, document requirements, assessment procedures, and coordination among personnel and government offices. However, the actual process used by the selected BPL-LIEO in Rosario, La Union may differ from the procedures implemented by other municipalities. Therefore, the present study uses the selected BPL-LIEO's actual workflow as the primary basis for system requirements.

Based on the information gathered for the present study, clients of the selected BPL-LIEO may initially approach personnel for inquiry, submission of requirements, or assistance in obtaining a queue ticket. The first service window plays an important receiving role because personnel assigned to the window accommodate clients and receive applications. The queue is currently monitored and managed by employees through computer-based information, while clients can monitor their queue position through the available display. When a client's number is called, the employee provides the instruction for the client to proceed to the appropriate service point.

The existing situation also indicates that new and renewal transactions may currently pass through similar stages even though their requirements can differ. A new business permit application requires the submission of the required documents and completion of applicable forms. In contrast, a renewal transaction may follow a simpler procedure when there are no changes in important business details such as ownership, line of business, business name, or business address. In such cases, existing information may be used and the transaction may proceed toward assessment after the required forms and approvals have been completed. This difference is important in designing the proposed queue management system because the selected transaction type should help determine the appropriate queue and service process.

The BPL-LIEO also requires clients processing regular permit transactions to accomplish a security checklist that serves as an indicator related to competitiveness. Consequently, the queue system should be designed around the actual workflow and requirements of the office instead of treating all clients as having exactly the same service process.

The literature and local context reveal several important points. First, queue management systems can improve the organization of clients and service resources. Second, real-time queue information can improve transparency and help clients determine when they should proceed for service. Third, priority mechanisms can allow personnel to handle exceptional queue conditions. Fourth, Philippine and regional government offices are increasingly adopting digital business permitting and queueing technologies. However, the existing literature and implementations reviewed do not establish a single queue

management system that exactly matches the operational requirements of the selected BPL-LIEO in Rosario, La Union.

Thus, the gap addressed by the present study is not simply the absence of a digital queueing system. Rather, the study seeks to develop a queue management system that is specifically adapted to the actual BPL-LIEO workflow, including transaction-based queue identification, real-time queue monitoring, staff-controlled queue operations, handling of absent clients, return-tomorrow transactions, priority check-in of returning clients, and queue history and statistics. The system is therefore intended to provide a practical solution that connects the client-facing queue process with the actual operational needs of BPL-LIEO personnel.

Statement of Objectives

This study aims to design and implement a Real-Time LAN-Based Queuing Management and Statistical Reporting System for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) to improve client flow, organize queue operations, and support efficient service delivery through data-driven insights.

Specifically, the study seeks to achieve the following objectives:

1. To design and develop an automated LAN-based queue management system that includes:
 - 1.1. A staff-facing ticket generator that prints a unique queue number based on the client's selected transaction type.

- 1.2. A staff-controlled dashboard that allows authorized personnel to call the next client, complete a transaction, skip a ticket, and manage priority/returning clients.
- 1.3. A real-time public monitor display that shows the current queue number and the next client to be served.
- 1.4. A statistical reporting module for administrators to track office trends and employee performance metrics.
2. To test and improve the system's performance, real-time data synchronization, and overall functionality through iterative alpha and beta testing in a local area network environment.
3. To evaluate the acceptability of the developed system in terms of functional suitability, usability, reliability, performance efficiency, security, maintainability, and portability based on the ISO 25010 Software Quality Standards, as assessed by the BPL-LIEO staff and clients.

Time and Place of Study

This study will be conducted at the Don Mariano Marcos Memorial State University – South La Union Campus, College of Computer Science, located in Consolacion, Agoo, La Union, and the Municipality of Rosario, specifically at the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO), Rosario, La Union for the School Year 2026–2027.

Definition of Terms

For a better understanding of this research report, the following terms are operationally defined:

Acceptability refers to the degree to which the intended users consider the developed queue management system understandable and suitable for supporting BPL-LIEO queue operations. This is measured through the following indicators: functional suitability, usability, reliability, performance efficiency, security, maintainability, and portability based on the ISO/IEC 25010 Software Quality Model.

Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) refers to the local government office of LGU Rosario, La Union responsible for business registrations, permit renewals, investment promotions, and licensing-related transactions. It serves as the primary beneficiary and implementation site of the proposed queue management system.

Client refers to an individual or business representative who visits the BPL-LIEO to inquire, submit requirements, process a new business permit, renew an existing permit, or perform another supported business-related transaction.

Current Queue Number refers to the queue number presently being served by the assigned service window. It is displayed in real time to inform waiting clients.

Done refers to the queue status assigned to a ticket when the client's transaction has been completed by the assigned personnel.

Monitor Display refers to the public-facing screen used to display the currently served queue ticket and succeeding queue information so that waiting clients can determine when they should proceed for service.

Queue History refers to the collection of recorded queue transactions from the system maintained by authorized personnel for monitoring, statistics, reporting, and future reference.

Queue Management System refers to the automated system developed in this study that manages the issuance of queue numbers, monitors queue status in real time, and supports staff-controlled queue operations to improve service efficiency in the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO).

Queue Number refers to the unique number automatically generated by the system and assigned to a client according to the selected transaction type. It determines the client's order of service.

Queue Status refers to the current condition of a queue ticket within the proposed system. The statuses include Waiting, Serving, Done, Return Tomorrow, and Priority Waiting.

Real-Time Status Updates refer to the live display of queue information, including the current queue number, the next queue number, and changes in queue status as transactions are processed.

Service Window refers to the designated counter where authorized personnel process clients' business permit and licensing transactions after their queue numbers are called.

Serving refers to the queue status assigned to a ticket when the client is currently being accommodated or processed by the assigned personnel.

Staff-Controlled Queue Operations refer to the functions available to authorized personnel for managing the queue. These include calling the next queue number, skipping, removing, and completing queue transactions to maintain an organized client flow.

Transaction Type refers to the category of service selected by the client, such as new business registration, business permit renewal, or other licensing-related transactions. It determines the appropriate queue assignment and service process.

Usability refers to the extent to which the queue management system can be effectively, efficiently, and satisfactorily used by its intended users. In this study, usability forms part of the evaluation based on the ISO/IEC 25010 Software Quality Model.

Waiting refers to the queue status assigned to a newly generated ticket that has not yet been called by authorized personnel.

Chapter 2

METHODOLOGY

Research Design

The researchers will utilize a combination of developmental research and descriptive research designs. Developmental research will be employed to systematically design, develop, and evaluate the Real-Time LAN-Based Queuing Management and Statistical Reporting System for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO). This approach is appropriate as the primary goal of the study is to create a functional technological solution to an existing problem.

Simultaneously, descriptive research will be used during the evaluation phase to assess the system's acceptability. This will involve surveying the end-users (BPL-LIEO staff and citizens) to describe their experience and gather quantitative feedback on the system's performance compared to the traditional manual queueing method.

Materials and Procedures

The following hardware and software tools will be used in the development of the system:

Hardware

- Laptops/Desktops for development and staff usage, a standard printer for ticket generation, a Smart TV or monitor for the queue display, and local area network (LAN) routers.

Software

- Frontend: React.js, HTML5, Vanilla CSS for the user interfaces (Staff Dashboard, Admin Panel, TV Display),
- Backend: Node.js with Express.js framework for handling API requests,
- Database: Prisma ORM with a relational database (SQLite/PostgreSQL) to store ticket records, user accounts, and analytics,
- Real-time Communication: Socket.io for live updates between the Staff Dashboard and the TV Display,
- Development Tools: Visual Studio Code (VS Code), Git for version control, and web browsers (Chrome/Edge) for testing.

Software Development Life Cycle (SDLC)

The study will adopt the **Agile Methodology** as its software development life cycle model. In Agile development, software is constructed incrementally through iterative cycles (sprints), allowing the development team to adapt to changing requirements, continuously integrate feedback, and systematically test features. The project will execute the following Agile phases:

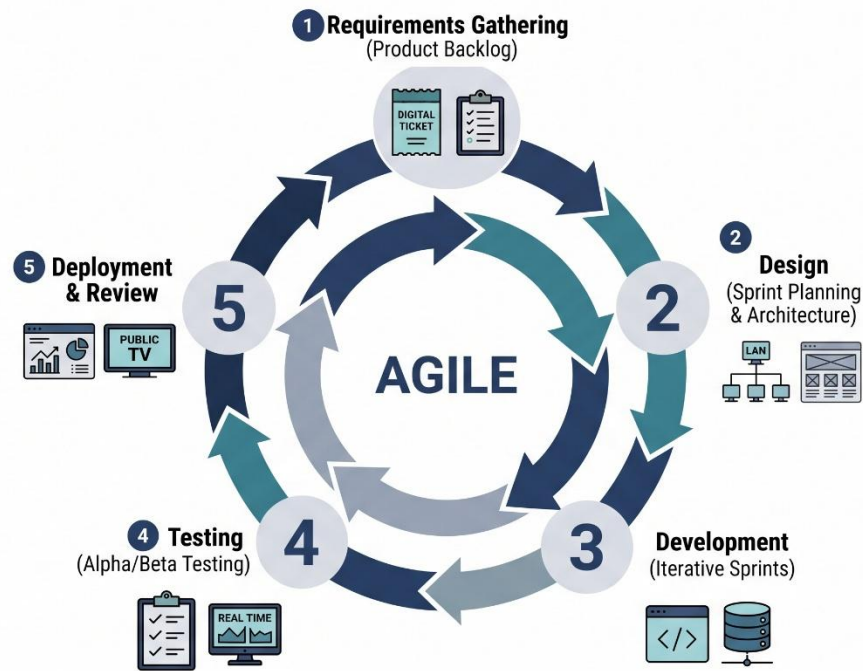


Figure 1. Custom Agile Software Development Life Cycle Model for the BPL-LIEO Queuing System

Requirements Gathering. (Product Backlog Creation). In Agile, this phase involves identifying user needs, defining project scope, and translating requirements into prioritized user stories that form the Product Backlog. The researchers conducted stakeholder interviews and observational workflow analysis with BPL-LIEO personnel to identify operational bottlenecks in manual queuing. Key functional requirements were established, including transaction-based ticket issuance, live counter calling, skipping/deferral handling, priority management for returning clients, and statistical reporting.

Design (Sprint Planning & Architectural Modeling). In the Agile framework, the design phase decomposes high-level backlog items into actionable sprint plans, user interface wireframes, and system architecture blueprints. The researchers designed the system architecture using Entity-Relationship Diagrams (ERD) for relational data persistence, Data Flow Diagrams (DFD Levels 0 and 1) for information routing across the network, Use-Case Diagrams for actor-role boundaries, and interactive wireframes for the Staff Dashboard, TV Display, and Admin Analytics panels.

Development (Iterative Sprints & Incremental Coding). In Agile, development is executed in iterative time-boxed sprints where functional modules are built, integrated, and verified incrementally. The researchers utilized a modern JavaScript tech stack comprising React.js and vanilla CSS for the responsive frontend user interfaces, Node.js with the Express.js framework for backend RESTful APIs, Prisma ORM with SQLite for reliable local data persistence, and Socket.io for bidirectional, real-time event broadcasting between the staff dashboards and the public TV display.

Testing (Continuous Integration & Alpha/Beta Testing). Agile emphasizes continuous testing alongside development to ensure code quality, socket event stability, and immediate defect resolution. The researchers will perform unit testing on individual API endpoints, integration testing on WebSocket event emitters/listeners, and stress testing under simulated concurrent client loads. Furthermore, formal Alpha Testing will be conducted in a simulated lab environment by peer developers, followed by Beta Testing conducted with actual BPL-LIEO personnel within the target local area network (LAN) environment.

Deployment and Review (Incremental Release & Evaluation). In Agile, this final phase delivers a working software increment into the operational environment, followed by a sprint review and retrospective to evaluate user satisfaction and gather feedback for future iterations. The researchers will deploy the system on the dedicated local area network router of the Rosario BPL-LIEO. End-users (BPL-LIEO staff and visiting clients) will interact with the system during live trial runs and evaluate its quality and acceptability using a structured evaluation questionnaire based on the ISO/IEC 25010 Software Quality Model.

To evaluate the acceptability and software quality of the developed system, a customized survey questionnaire adapted from the ISO/IEC 25010 Software Quality Standards (see Appendix D) will be distributed to the respondents. The evaluation instrument assesses the following seven (7) quality characteristics:

1. **Functional Suitability** – assesses the degree to which the system provides functions that meet stated transaction and queuing requirements (functional completeness, correctness, and appropriateness).
2. **Usability** – assesses how easily BPL-LIEO personnel and clients can understand, learn, and operate the user interfaces and monitor displays.
3. **Performance Efficiency** – assesses the response times, transaction throughput, and resource utilization of the system during real-time socket communications across the LAN.
4. **Reliability** – assesses the system's fault tolerance, data consistency, and ability to recover gracefully from network interruptions.

5. **Security** – assesses counter authentication, role-based access control, and the protection of transaction logs against unauthorized tampering.
6. **Maintainability** – assesses the modularity, testability, and modifiability of the codebase using Prisma schemas and Express controllers.
7. **Portability** – assesses the ease with which the system can be deployed and executed across different client devices, browsers, and standard LAN configurations.

The respondents of the study are the staff of the BPL-LIEO and citizens/clients. The researchers will distribute the survey questionnaire to two distinct user groups: all BPL-LIEO staff members who operate the ticket generator and staff dashboard, and visiting citizens/clients who receive printed tickets and observe the public TV display. Because the visiting client population fluctuates daily, convenience sampling ($n = 50$) will be utilized for citizens, while total enumeration ($N = 10$) will be employed for the BPL-LIEO staff.

Table 1. Distribution of Respondents

Respondents	Total Population	Sample Population
	N	n
BPL-LIEO Staff	10	10
Citizens / Clients	150	50
TOTAL	160	60

System Testing Procedures

To present the results for testing and improving the system (Specific Objective 2), this section will feature the following outputs upon the completion of the study:

- **Summary of Alpha and Beta Testing Results:** A matrix listing the core functionalities tested (e.g., Ticket Generation, Calling Next Client, Real-time TV Display Update, Skipping Client), showing the expected output versus the actual output, and the pass/fail status of each module.
- **Defect Resolution and System Improvements Report:** A detailed narrative detailing the technical issues encountered during the Alpha and Beta testing phases (such as network latency or socket disconnections) and the corresponding code optimizations applied by the researchers to resolve them.
- **Real-Time Synchronization Performance:** An analysis of how the system performed over the BPL-LIEO's local area network, validating that real-time data broadcasting was successfully achieved without reliance on an external internet connection.

Data Gathered

The researchers will gather information on the current state of service delivery and workflow in the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO). The system's acceptability will also be gathered through a survey using the ISO-Based questionnaire to both the BPL-LIEO staff and clients.

Data Analysis

To analyze the data gathered from the survey questionnaires, the formula for the weighted mean will be used to interpret the overall acceptability of the queuing management and statistical reporting system. This statistical tool is appropriate for determining the average response of the participants based on a 5-point Likert Scale. Below presents the average score range and the corresponding verbal interpretation.

Scale	Mean Score	Descriptive Equivalent Rating
1 (Strongly Disagree)	1.00 – 1.79	Poor
2 (Disagree)	1.80 – 2.59	Fair
3 (Neutral)	2.60 – 3.39	Good
4 (Agree)	3.40 – 4.19	Very Good
5 (Strongly Agree)	4.20 – 5.00	Excellent

Responses with a score ranging from 3.40 to 5.00 indicate varying levels of agreement. Specifically, scores between 4.20 and 5.00 (Strongly Agree) indicated that the system's quality is excellent. Scores between 3.40 and 4.19 (Agree) suggested that the website quality is very good, while scores between 2.60 and 3.39 (Neutral) indicated that the system's quality is good. Conversely, responses falling below 2.60, signifies client dissatisfaction.

LITERATURE CITED

- Ainbinder, I., Temnikov, E., & Allalouf, M. (2024). A study comparing waiting times in global and local queuing systems with heterogeneous workers. *Applied Sciences*, *14*(9), 3799. <https://doi.org/10.3390/app14093799>
- Alegado, R., & Himo, A. J. (2022). Community request queue management system for Local Government Unit of Cabanatuan City. *International Journal of Advanced Trends in Computer Science and Engineering*, *11*(6), 285–293. <https://doi.org/10.30534/ijatcse/2022/091162022>
- Anti-Red Tape Authority. (2024, January 18). *Interagency efforts key to improved eBOSS implementation*. <https://pia.gov.ph/press-release/interagency-efforts-key-to-improved-eboss-implementation/>
- Chantima, S., & Pochan, J. (2023). Real-time queue management based on equality to reduce waiting time by using mathematical model. *Social Science Journal of Prachachuen Research Network*, *5*(2), 84–98. <https://so03.tci-thaijo.org/index.php/prn/article/view/267615>
- Dela Merced, C. L., Balmeo, B. E., & De Castro, L. B. (2024). Exploring client perspective: A case study of the Electronic Business One-Stop Shop (E-BOSS) towards client satisfaction for SMEs in Quezon City. *Journal of Emerging Technologies and Innovative Research*, *11*(4), 516–524.

<https://neuresearchcenter.com/2025/08/29/exploring-client-perspective-a-case-study-of-the-electronic-business-one-stop-shop-e-boss-towards-client-satisfaction-for-smes-in-quezon-city/>

Espinosa, G. S., Serrano, M. R., & Balmes, H. A. (2024). Automated enrollment queueing system in a university towards student experience and operational efficiency. *MSEUF Research Studies*, 21(1), 71–77.

<https://ejournals.ph/article.php?id=24562>

Fauzi, N. A. b. M., & Suhaimi, N. H. S. (2026). Design and evaluation of QKlinik: A web-based queue management system for outpatient clinics. *International Journal of Software Engineering and Computer Systems*, 11(2), 147–159.

<https://doi.org/10.15282/ijsecs.11.2.2025.12.0144>

International Organization for Standardization. (2011). *Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — System and software quality models* (ISO/IEC Standard No. 25010:2011).

<https://www.iso.org/standard/35733.html>

Lee, S., Dudin, A., Dudina, O., & Kim, C. (2024). Analysis of a priority queueing system with the enhanced fairness of servers scheduling. *Journal of Ambient Intelligence and Humanized Computing*, 15, 465–477. [https://doi.org/10.1007/s12652-022-](https://doi.org/10.1007/s12652-022-03903-z)

03903-z

Local Government Unit of Agoo, La Union. (2025). *Agoo's Business One-Stop Shop (BOSS) for C.Y. 2025*. <https://agoolaunion.gov.ph/Agoos-Business-One/>

Municipality of Sudipen, La Union. (2025). *Business One-Stop Shop*.
<https://www.sudipenlaunion.gov.ph/business-one-stop-shop/>

Philippine Information Agency. (2024, April 25). *ARTA commends La Union town's eBOSS implementation*. <https://pia.gov.ph/news/arta-commends-la-union-towns-eboss-implementation/>

Provincial Government of La Union. (n.d.). *Office of the Provincial Treasurer*.
<https://launion.gov.ph/department/office-of-the-provincial-treasurer/>

APPENDICES

APPENDIX A

Letter to the Adviser

APPENDIX B

Interview Guide Questions

APPENDIX C

Evaluation Questionnaire

Name (Optional): _____

Date: _____

User Type: ☐ BPL-LIEO Staff ☐ Citizen / Client

Instructions: Please evaluate the Real-Time LAN-Based Queuing Management and Statistical Reporting System by checking the box that corresponds to your level of agreement with the following statements.

Scale:

5 - Strongly Agree (Excellent)

4 - Agree (Very Good)

3 - Neutral (Good)

2 - Disagree (Fair)

1 - Strongly Disagree (Poor)

Criteria	5	4	3	2	1
I. Functional Suitability					
1. The system generates accurate queue numbers based on the selected transaction type (New vs. Renewal).					
2. The staff dashboard provides all the necessary functions (Call, Skip, Done, Return Tomorrow) to manage the queue.					
3. The statistical reporting module accurately tracks and summarizes daily queue transactions.					

II. Usability					
1. The public monitor display (TV) is easy to read and understand from a distance.					
2. The staff interface is user-friendly and easy to navigate without extensive training.					
3. The printed queue ticket provides clear and helpful information to the client.					
III. Performance Efficiency					
1. The system responds quickly when generating a new ticket.					
2. The public monitor updates in real-time (without noticeable delay) when a staff member calls the next number.					
3. The system operates smoothly even when multiple clients are queuing simultaneously.					
IV. Reliability					
1. The system operates without unexpected crashes or freezing during operation.					
2. The system retains accurate queue data even if the local network experiences temporary interruptions.					
V. Security					
1. The system requires proper authentication (login) for staff members to access the dashboard.					
2. Only authorized personnel can modify queue statuses or access the statistical reports.					
VI. Maintainability					
1. The system's settings (e.g., adding a new transaction type or service window) are easy to update.					
2. The statistical records are easy to retrieve and export for office audits.					
VII. Portability					

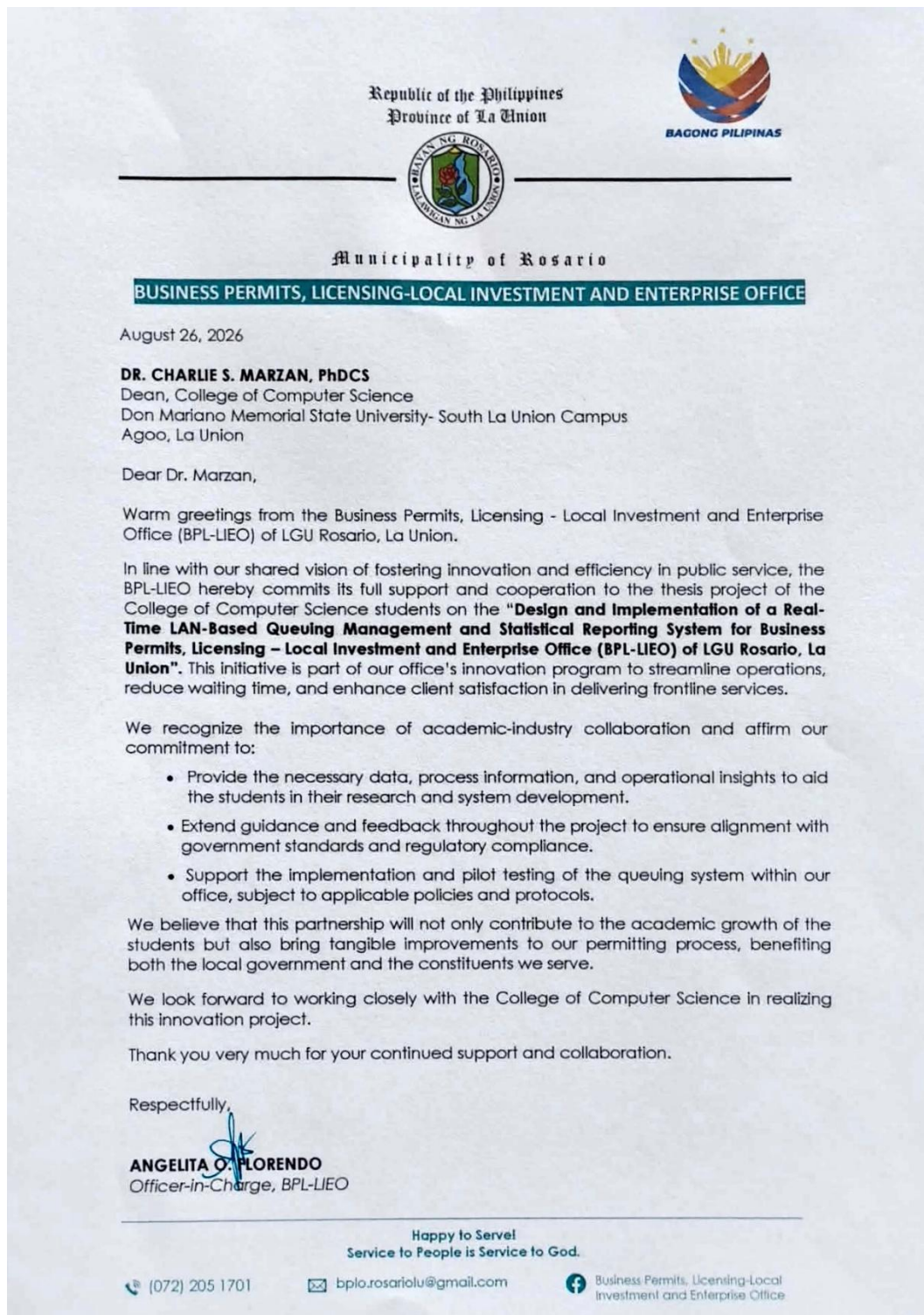
1. The system can be accessed easily through different web browsers (e.g., Chrome, Edge).					
2. The staff dashboard scales properly when viewed on different laptop or desktop screen sizes.					

APPENDIX D

Letter to the Panel Members

APPENDIX E

Commitment Letter



CURRICULUM VITAE

ROJEAN F. TANDAS

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rtandas23109793@student.dmmmsu.edu.ph



Personal Information

Birthdate: July 22, 2005

Place of Birth: Rosario, La Union

Gender: Male

Age: 21

Nationality: Filipino

Religion: Roman Catholic

Civil status: Single

Father's Name: Bencio M. Tandas

Mother's Name: Maribel F. Tandas

Educational Background

2023-present

College Level

Bachelor of Science in Computer Science

Don Mariano Marcos Memorial State University

Consolacion, Agoo, La Union

Design and Implementation of a Real-Time LAN-Based Queuing Management and Statistical Reporting System for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) of LGU Rosario, La Union

2021-2023

Senior High School Level

Santo Tomas National High School

Patac, Santo Tomas, La Union

With Honors

2016-2021

Junior High School Level

Santo Tomas National High School
Patac, Santo Tomas, La Union

2011-2016

Elementary Level

Casilagan Elementary School
Casilagan, Santo Tomas, La Union

Special Skills and Talents

Technical Skills

- Programming Languages: C, JavaScript
- Web Technologies: HTML / HTML5
- System & Environment Management: Software Installation, OS Configuration, Basic Troubleshooting
- Productivity Software: Microsoft 365 (Word, Excel, PowerPoint)

Capabilities, Strengths & Core Aptitudes

- Technical Troubleshooting: Ability to configure system environments, install essential developer tools, and resolve baseline software issues.
- Foundational Algorithmic Thinking: Understanding of procedural programming concepts, control structures, and syntax logic via C.
- Front-End Development Basics: Structuring clean, semantic web layouts and implementing foundational client-side scripting.
- Documentation & Reporting: Creating structured technical write-ups, presentations, and spreadsheet records.

I hereby certify that the above statements are correct to the best for my knowledge.


ROJEAN F. TANDAS

Signature over printed name

CARREN NICOLE A. ESTRELLES

Namonitan, Santo Tomas, La Union

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cnestrelles22111243@student.dmmmsu.edu.ph



Personal Information

Birthdate: August 21, 2004

Place of Birth: Santo Tomas, La Union

Gender: Female

Age: 22

Nationality: Filipino

Religion: Born Again Christian

Civil status: Single

Father's Name: Edward C. Estrelles

Mother's Name: Marites A. Estrelles

Educational Background

2022-present

College Level

Bachelor of Science in Computer Science

Don Mariano Marcos Memorial State University

Consolacion, Agoo, La Union

Design and Implementation of a Real-Time LAN-Based Queuing Management and Statistical Reporting System for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) of LGU Rosario, La Union

2020-2022

Senior High School Level

Polytechnic College of La Union

San Nicolas Sur, Agoo, La Union

2016-2020

Junior High School Level

Polytechnic College of La Union

San Nicolas Sur, Agoo, La Union

With Honors

2010-2016

Elementary Level

Fort Del Pilar Elementary School
Philippine Military Academy, Baguio City

Special Skills and Talents

Technical Skills

- Web Development: HTML5, CSS3, JavaScript (ES6)
- Hardware & Systems: Computer Hardware Diagnostics, Component Assembly, System Maintenance & Upgrades

Capabilities, Strengths & Core Aptitudes

- Front-End Fundamentals: Developing structured, styled, and interactive user interfaces using modern client-side web technologies.
- Hardware Troubleshooting: Identifying component-level faults, handling peripheral integration, and performing routine hardware maintenance.
- Systems Familiarity: Understanding the interplay between physical hardware architecture and software execution.

I hereby certify that the above statements are correct to the best of my knowledge.



CARREN NICOLE A. ESTRELLES

Signature over printed name

JADE ALLYSA V. RULLODA

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Personal Information

Birthdate: September 20, 2004

Place of Birth: San Fernando, La Union

Gender: Female

Age: 21

Nationality: Filipino

Religion: Roman Catholic

Civil status: Single

Father's Name: Wilson D. Rulloda

Mother's Name: Susan V. Rulloda

Educational Background

2023-present

College Level

Bachelor of Science in Computer Science

Don Mariano Marcos Memorial State University

Consolacion, Agoo, La Union

Design and Implementation of a Real-Time LAN-Based Queuing Management and Statistical Reporting System for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) of LGU Rosario, La Union

2021-2023

Senior High School Level

Aringay National High School

Poblacion, Aringay, La Union

With Honors

2017-2022

Junior High School Level

Aringay National High School

Poblacion, Aringay, La Union

2011-2016

With Honors

Elementary Level

Aringay Central Elementary School
Poblacion, Aringay, La Union

Special Skills and Talents

Technical Skills

- Programming Languages: Java
- Web Technologies: HTML / HTML5
- Core Computing: Environment Setup, File Management, System Operations

Capabilities, Strengths & Core Aptitudes

- Object-Oriented Programming (OOP) Fundamentals: Understanding foundational OOP principles, class structures, and programmatic logic in Java.
- Web Content Structuring: Building semantic, accessible document outlines and baseline web structures using HTML.
- Technical Adaptability: Quickly learning syntax structures, core computing workflows, and essential developer environments.

I hereby certify that the above statements are correct to the best for my knowledge.


JADE ALLYSA V. RULLODA
Signature over printed name

JOHN DAVID R. GARCIA

Tocok, San Fabian, Pangasinan

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Personal Information

Birthdate: November 3, 2004

Place of Birth: Dagupan City, Pangasinan

Gender: Male

Age: 21

Nationality: Filipino

Religion: Born Again Christian

Civil status: Single

Father's Name: Victoriano S. Garcia

Mother's Name: Marites R. Garcia

Educational Background

2023-present

College Level

Bachelor of Science in Computer Science

Don Mariano Marcos Memorial State University

Consolacion, Agoo, La Union

Design and Implementation of a Real-Time LAN-Based Queuing Management and Statistical Reporting System for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) of LGU Rosario, La Union

2021-2023

Senior High School Level

Graystone Institute of the Philippines

3F Maradel Bldg., AB Fernandez Street,

Dagupan City, Pangasinan

With High Honors

2016-2021

Junior High School Level

San Fabian National High School

Cacapit Street, San Fabian, Pangasinan

With Honors

2011-2016

Elementary Level

Tocok Elementary School
Tocok, San Fabian, Pangasinan
With Honors

Special Skills and Talents

Technical Skills

- Languages & Web Stack: Python, Java, JavaScript, React, Node.js, Express, Tailwind CSS, Semantic HTML/CSS.
- Systems & Environment: Linux (Ubuntu) Administration, Git/GitHub, Supabase, CLI Operations.
- Architecture & Design: UML Modeling, MVP Feature Scoping, Figma (UI Mockups & Structural Site Maps).

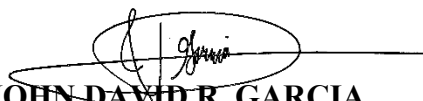
Capabilities, Strengths & Core Aptitudes

- AI Prompt Engineering: Crafting highly specific master prompts, custom instructions, and behavioral parameters to optimize coding and technical workflows.
- System & OS Optimization: Diagnosing and safely debloating mobile operating systems via precise system package management.
- Product & Business Strategy: Formulating Business Model Canvases, transaction flows, and operational workflows for conceptual software platforms.

Leadership & Soft Skills

- Public & Academic Governance: Managing public funds and community initiatives as an SK Treasurer, alongside logistical coordination and peer facilitation as a Class Secretary and Captain.
- Technical Communication: Translating system architectures into formal client proposals and delivering structured presentations on programming philosophies.

I hereby certify that the above statements are correct to the best for my knowledge.


JOHN DAVID R. GARCIA
Signature over printed name

MANRIJAN M. BARBOSA

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mbarbosa2310980@student.dmmmsu.edu.ph



Personal Information

Birthdate: June 19, 2004

Place of Birth: Baguio City

Gender: Male

Age: 22

Nationality: Filipino

Religion: Roman Catholic

Civil status: Single

Father's Name: Manrigold E.Barbosa

Mother's Name: Florence M. Barbosa

Educational Background

2023-present

College Level

Bachelor of Science in Computer Science

Don Mariano Marcos Memorial State University

Consolacion, Agoo, La Union

Design and Implementation of a Real-Time LAN-Based Queuing Management and Statistical Reporting System for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) of LGU Rosario, La Union

2020-2022

Senior High School Level

Rosario Integrated School

Poblacion East Rosario La Union

2016-2020

Junior High School Level

Saint Mary's Academy

Poblacion East Rosario La Union

2010-2016

Elementary Level

Rosario Integrated School

Poblacion East Rosario La Union

Special Skills and Talents

Technical Skills

- Software Development: Core Programming Fundamentals, Code Logic, and Syntax Application
- Data & Analytics: Data Analysis, Information Processing, and Analytical Modeling
- Design & Multimedia: Graphic Design, Digital Asset Creation, and Visual Communication

Capabilities, Strengths & Core Aptitudes

- Analytical Problem-Solving: Applying structured methodologies to identify, assess, and resolve complex technical and logistical challenges.
- Technical Troubleshooting: Diagnosing system errors, debugging code logic, and implementing effective, practical solutions.
- Dynamic Adaptability: Quickly acclimating to new technologies, development environments, and shifting project requirements.
- Effective Communication: Articulating technical concepts clearly to both technical peers and cross-functional team members.

I hereby certify that the above statements are correct to the best for my knowledge.

MANRIJAN M. BARBOSA

Signature over printed name